MILITARY SPECIFICATION

SHAKING APPARATUS, LABORATORY (ROTATING TYPE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for the use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers slide rotating apparatus for simulating hand shaking of micro slides, antigen flasks, etc., for flocculation tests in connection with cardiolipin syphilis tests.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 —Boxes, Wood, Nailed and Lock-Corner.

LLL-B-631 —Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL-B-636 —Boxes; Fiber, Solid (for Domestic Shipment).

MILITARY

JAN-P-106 —Packaging and Packing for Overseas Shipment — Boxes; *Wood*, Nailed.

JAN-P-125 —Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN-P-140 —Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Material shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

3.1.1 *Platform.*—Platform shall be of hard or half-hard cold rolled steel, covered with industrial sponge rubber.

3.1.2 *Springs.*—Springs shall be of corrosion-resisting steel.

3.1.3 *Base.*—Base shall be of cold-or hot-rolled steel channel, coated with black asphalt varnish.

3.1.4 *Housing or guard.*—Housing or guard shall be of hard or half-hard rolled steel sheet with protective, baked-on enamel finish.

3.2 Platform.—Platform shall be a rubber covered, floating platform, approximately 13 inches square, to accommodate approximately

983324—52—(42)

Source: https://assist.dla.mil -- Downloaded: 202610-16T21:12Z
Check the source to verify that this is the current version before use.

fifteen 57 by 108 mm. Boerner slides. Metal shall be approximately 1/8 inch thick and shall be securely, completely, and permanently covered on the top side with flat industrial sponge rubber, not less than 1/8 inch thick. Platform shall be attached to four corrosion-resisting, vertical springs, one near each corner of the bottom side, by means of corrosion-resisting nuts and bolts. The opposite end of each spring shall be secured near the corners of the metal base by similar nuts and bolts. Springs shall be made of approximately 1/8 inch diameter wire, shall have approximately nine loops of approximately 1 1/8 inches outer diameter, and shall be tapered or shaped at the ends to accommodate nuts and bolts securely. Platform shall be supplied with four retaining bands of rubber for holding the slides to prevent creeping when rotator is in operation. Rubber covered top of platform shall present a coarse surface to minimize slippage of slides.

3.3 **Base.**—Base shall be a steel channel 12 inches wide by 2 1/4 inches high by 1/4 inch thick and 12 inches long, approximate, all dimensions.

3.4 **Housing.**—Housing shall be a metal strip approximately 5 inches wide to engage the flanges of the base by means of suitable nuts and bolts. It shall enclose three sides and approximately two-thirds of back side so that sufficient area is exposed for accessibility to the interior of the apparatus. The front side, bearing controls shall be secured to the base by means of a metal angle, which shall be attached on the inside.

3.5 **Motor.**—Motor shall be special, continuous duty, 115 volt, a.c.-d.c., universal motor, 1/30 hp., with worm reduction gear. Worm and gear driving mechanism shall rotate the floating platform as specified in 3.11.1 to simulate hand rotation. Universal crank connection to the floating platform shall be made by means of a standard ball bearing, mechanically attached to the platform with shockproof mounting, and shall require no oiling for maintenance.

3.6 **Controls.**—Controls shall be permanently mounted on the front side of the housing.

3.6.1 *"Off-On" switch.*—"Off-On" switch shall be a suitable capacity toggle switch.

3.6.2 *Speed control rheostat.*—Speed control rheostat shall consist, in part, of an easily adjustable pointer on a suitably graduated dial. The rheostat shall control the speed of the platform rotation as specified in 3.11.2.

3.6.3 *Automatic cutoff timing device.*—Automatic cutoff timing device shall consist, in part, of an easily adjustable and securely mounted pointer on a dial which shall be graduated to indicate 0 to 28 minutes, in 1/2 minute subdivisions with at least every 5 minute interval numbered. Setting to a predetermined interval shall be accomplished by a simple adjustment of the pointer on the graduated dial. The apparatus shall operate continuously and independently of the automatic cutoff timing device when the toggle switch is in the "on" position. The apparatus shall operate on the timing device when the toggle switch is in the "off" position.

3.7 **Extension cord and plug.**—Extension cord and plug shall be a flexible rubber covered two-conductor cord, not less than 5 feet outside length, with a suitable rubber, 2-pole parallel blade, fingergrip attachment plug. Cord shall be secured in a manner to prevent strain on the terminal connections.

3.8 **Rubber cushions.**—Four solid or suction rubber cushions shall be secured, one at each corner of the base, to act as legs.

3.9 **Dimensions.**—Overall dimensions of apparatus shall be approximately 13 by 13 by 7 1/2 inches high.

3.10 **Service data.**—The contractor shall enclose with each unit two copies of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of spare parts. Wiring

Source: https://assist.dla.mil -- Downloaded: 20260-16T21:12Z
Check the source to verify that this is the current version before use.

diagrams shall be included for electrical items or components. Each part other than common fasteners, etc., shall be identified by the manufacturer's part number. In addition, the contractor shall mail one copy of the service data to Specifications Branch, Armed Services Medical Procurement Agency, 84 Sands Street, Brooklyn 1, N. Y., and one copy to Medical Technical Maintenance Division, Armed Services Medical Procurement Agency, St. Louis Medical Depot, 12th and Sprice Streets, St. Louis 2, Mo.

3.11 **Performance.**

3.11.1 *Platform rotation.*—Each point on platform shall rotate on a horizontal plane circling an area approximately ¾ inch in diameter to simulate hand rotation.

3.11.2 *Speed control.*—The apparatus shall be so regulated that when the pointer on the speed control rheostat is set at the lowest mark, the platform shall rotate approximately 70 times per minute. Rotation speed of the platform shall increase progressively with the adjustment of the pointer to the higher marks until at the maximum mark, the platform shall rotate approximately 210 times per minute.

3.12 **Workmanship.**—Workmanship shall be first class throughout. Rotator shall be free from defects which detract from its appearance or may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Samples shall be submitted as specified (see 6.1). In addition, inspection at contractor's facility by the Government inspector shall be conducted in accordance with Standard MIL–STD–105.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished articles shall be made either at point of production or at point of delivery, as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 **Tests.**—Tests shall be conducted to determine compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit packaging.*—One rotating apparatus constitutes one unit. Each unit shall be packaged in a solid fiberboard or double-faced corrugated fiberboard carton, of appropriate size and design, constructed in accordance with Specification LLL–B–631 or LLL–B–636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Each carton shall be adequately secured to prevent accidental opening.

5.2 Packing.

5.2.1 *For domestic shipment.*—Each unit carton (1 unit) shall be overpacked in an exterior container constructed in accordance with Specification NN–B–621.

5.2.2 *For overseas shipment.*—Each unit carton (1 unit) shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with Specification JAN–P–106. Closure and strapping shall comply with appendix thereto. Each box shall be lined with a sealed waterproof barrier (bag) fabricated of material conforming to Specification JAN–P–125. Closures shall be completely sealed by a continuous seam of water-resistant adhesive at least ¾ inch wide and conforming to Specification JAN–P–140.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:12Z
Check the source to verify that this is the current version before use.

5.3 **Marking.**—Shipping container shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Sampling required (see 4.1).

(c) Inspection required (see 4.2).

(d) Place of final inspection (see 4.2).

(e) Whether packing for domestic or overseas shipment is required (see 5.2).

6.2 This specification covers the following item appearing in the Armed Services Catalog of Medical Materiel:

4–389–600 Rotating Apparatus, Slide, 110 Volt, AC-DC.

6.3 This specification does not include all types, classes, grades, sizes etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Medical Service

Other interest:
Army—T
Navy—Md.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:12Z
Check the source to verify that this is the current version before use.